

Forensic ToolKit Lab

CyberDefenders Lab
Created June 9th, 2026

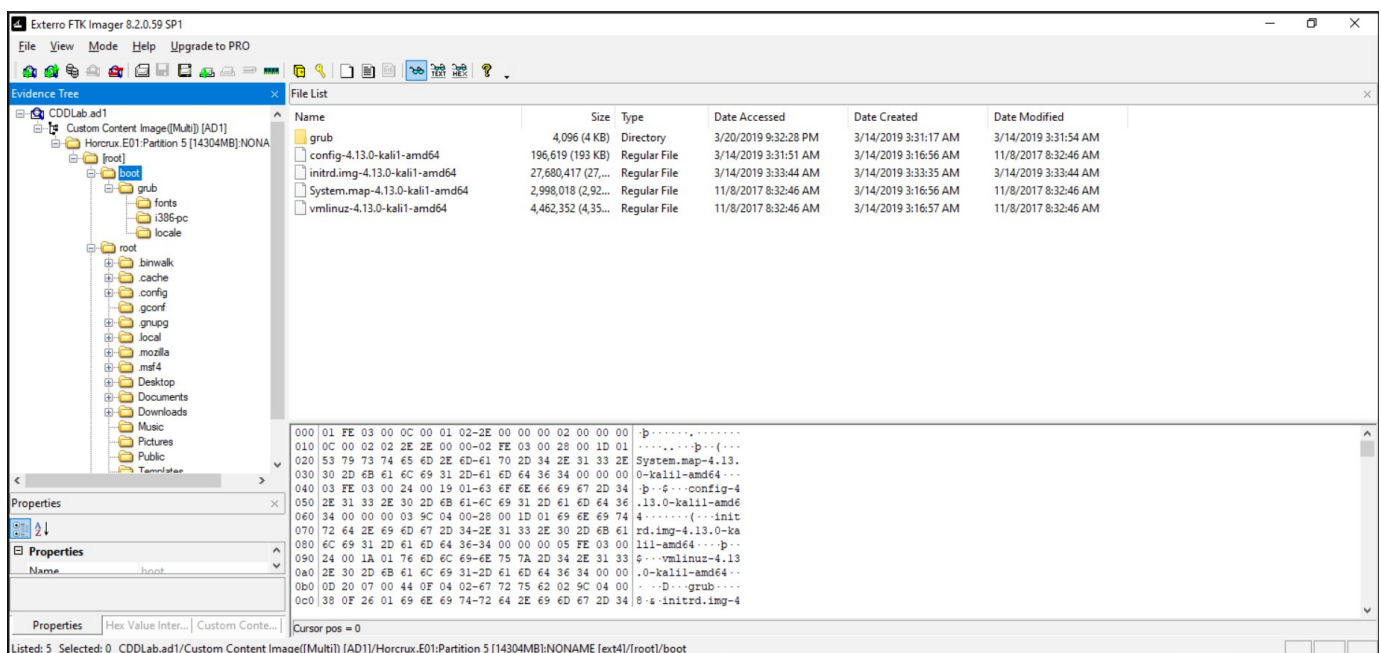
Scenario: After Karen started working for 'TAAUSAI,' she began doing illegal activities inside the company. 'TAAUSAI' hired me as a soc analyst to kick off an investigation on this case.

I acquired a disk image and found that Karen uses Linux OS on her machine. I must analyze the disk image of Karen's computer and answer the provided questions.

Prior to starting this lab, I needed to install FTK Imager. To do this, I went to the official website of Exterro, and installed FTK Imager 8.2.0.59 on my windows 10 VM. After the installation, I created a disk image of the evidence that was collected, and started to gather information for the questions.

Question 1: Which Linux distribution is being used on this machine?

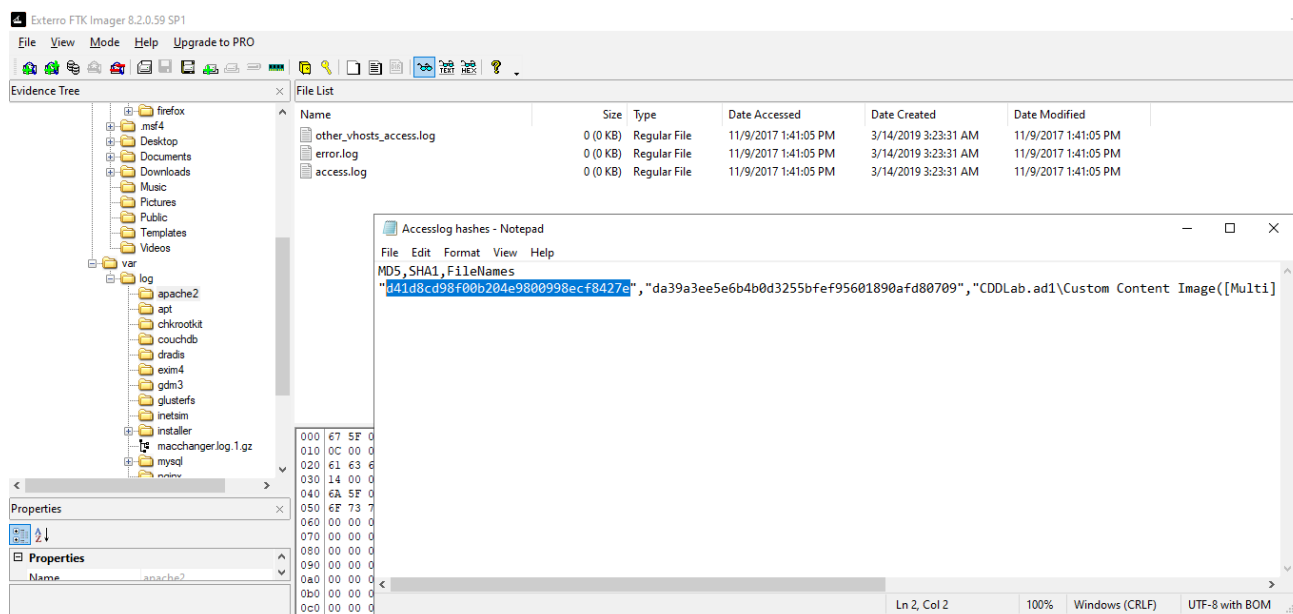
After collecting the disk image, I was then able to start navigating the file system of this image. To find out what Linux distribution was being used, I searched for the “boot” folder, since that would hold the information of the boot drive. From looking in there, I can see the Linux Distro is Kali



Question 2: What is the MD5 hash of the Apache access.log file?

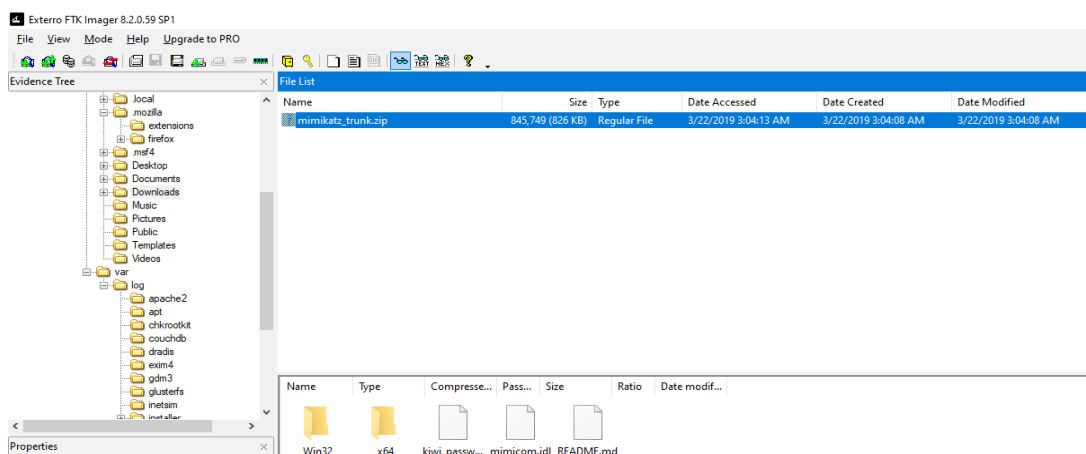
The Apache access.log text file is generated by an http server that records all requests processed by the web server. From here it details information such as who visited the website, what resources are requested, when the action occurred, and what the server response was.

To find the Apache access log, I have to navigate down to the var>log>apache2, which is where the access.log file was located. After that, I had to export it to receive the hashes in a text file, which included the md5 and sha1 hash.



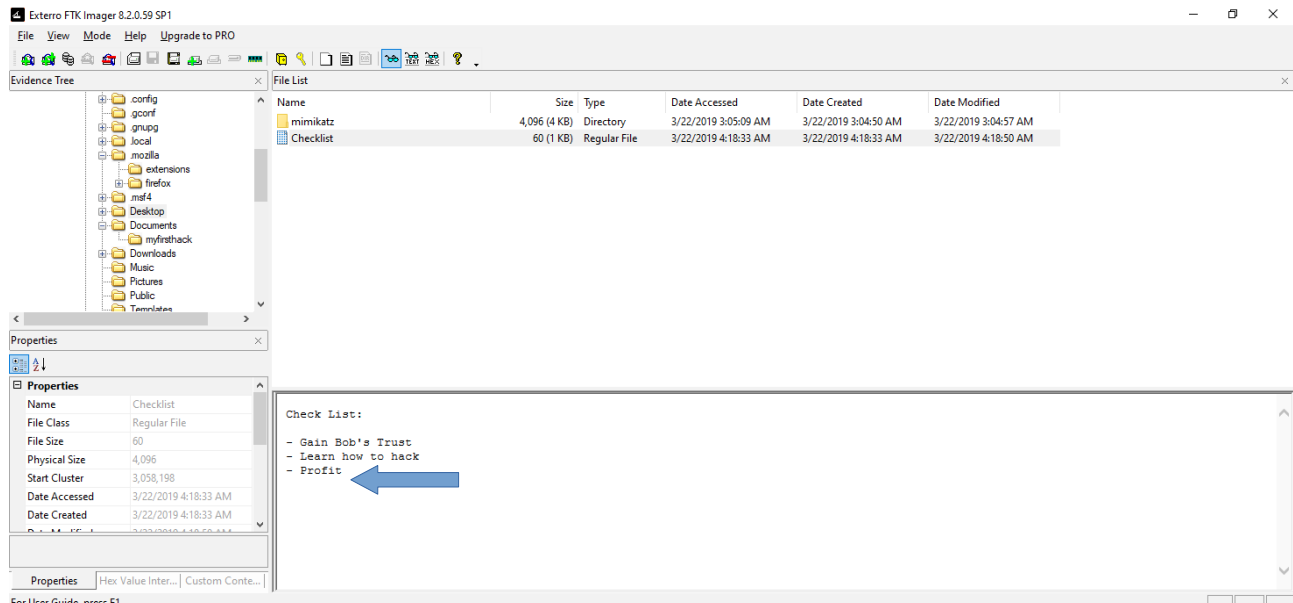
Question 3: It is suspected that a credential dumping tool was downloaded. What is the name of the downloaded file?

To find out what tool was used. I went into the downloads folder, and I found that the tool is mimikatz_trunk.zip. Mimikatz is a common credential dumping tool that is very powerful, and can be used for attacks such as pass-the-hash, pass-the-ticket, retrieving plaintext passwords, and many more attacks.

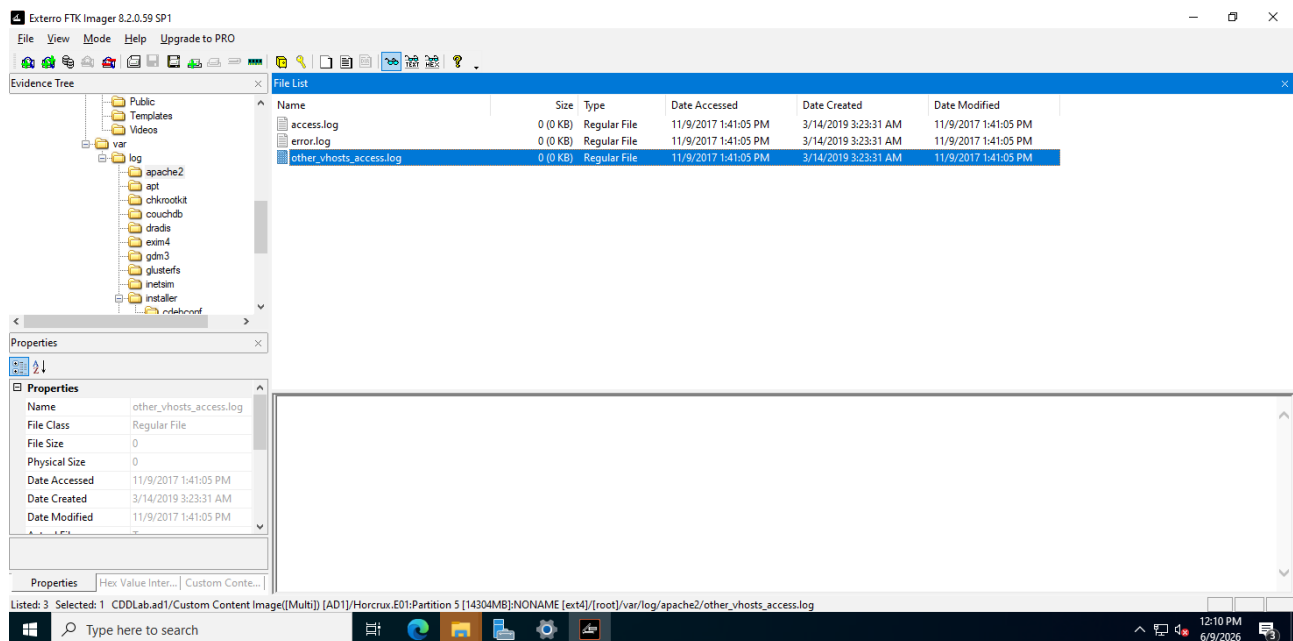


Question 6: What is the third goal from the checklist Karen created?

To find this checklist, just went to the desktop, found the checklist, and the answer is to profit



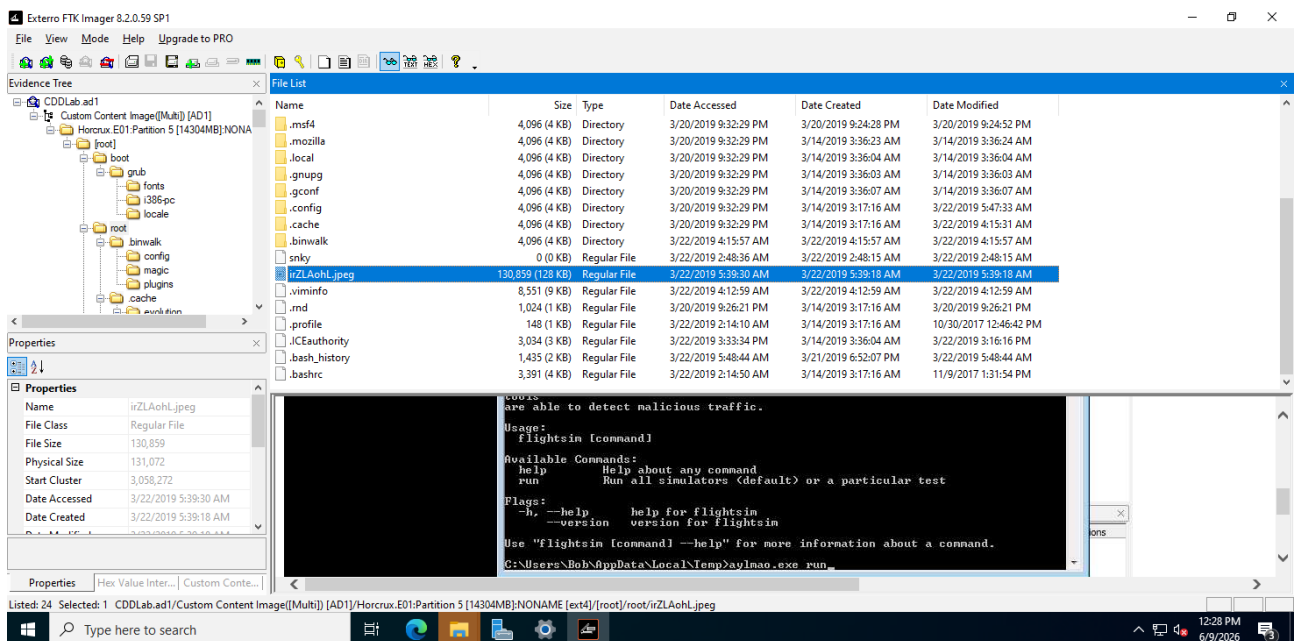
Question 7: How many times was Apache run?



When going back to `var>log>apache`, all the files inside are showing as empty, which shows that it was not ran.

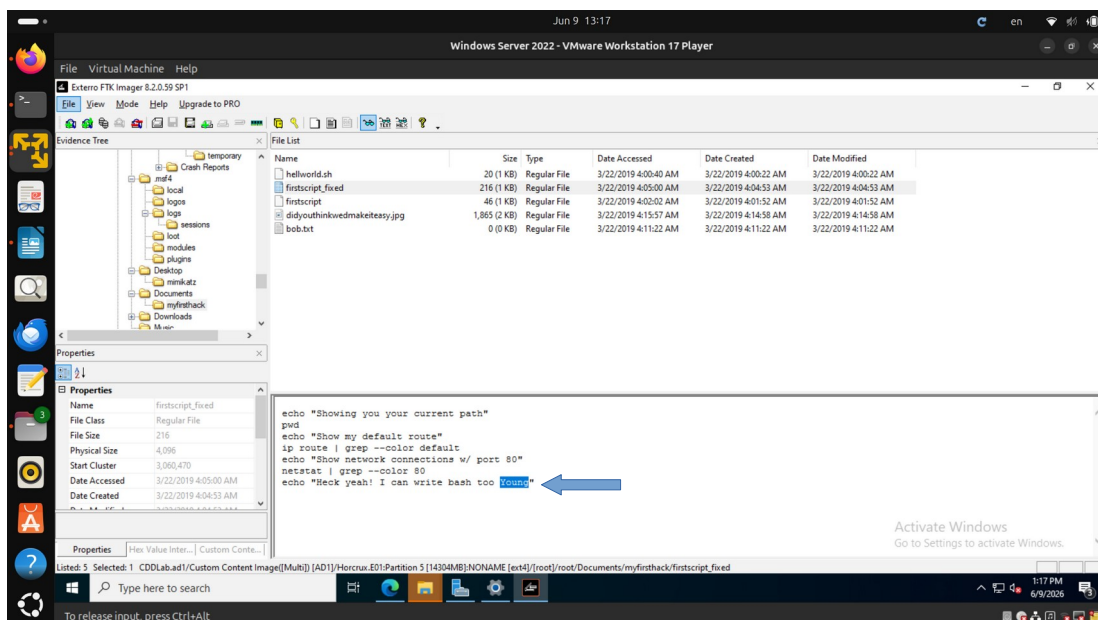
Question 8: This machine was used to launch an attack on another. Which file contains the evidence for this?

The evidence for this would be found in root under a jpeg image. we see a screenshot of a Windows desktop taken during an attack. This is where the information can be found.



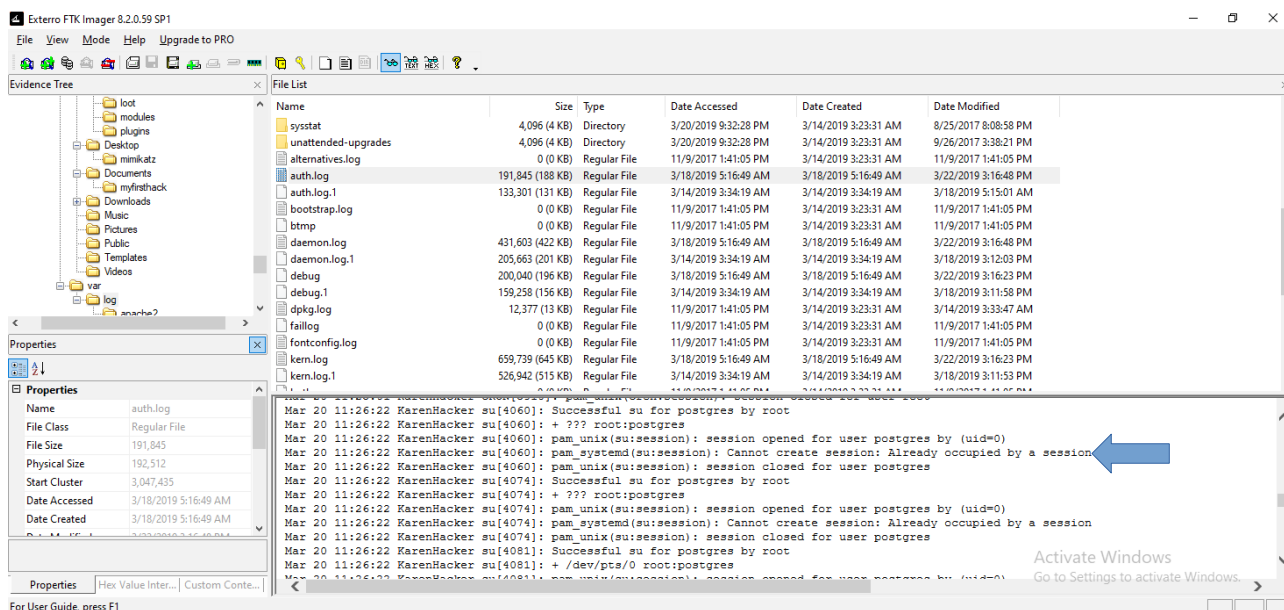
Question 9: It is believed that Karen was taunting a fellow computer expert through a bash script within the Documents directory. Who was the expert that Karen was taunting?

I navigated to the directory, checked one of the firsts scripts, and saw she is taunting Young.



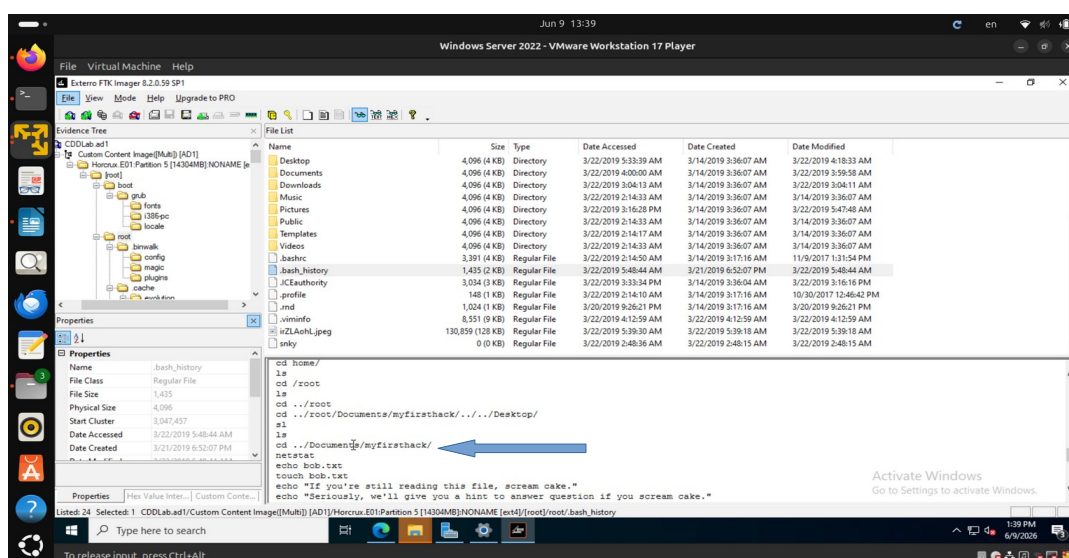
Question 10: A user executed the su command to gain root access multiple times at 11:26. Who was the user?

Since this is asking us to find who attempted access and what times, it is best to check the authentication logs. This can be found in var/logs/auth.log. Within this, I go to the time 11:26, and I can see that the user executed su commands for the “postgres” account.



Question 11: Based on the bash history, what is the current working directory?

From what I was able to find in the bash history, the user was constantly referring to the directory /root/documents/myfirsthack



Final Thoughts

Overall, it was a pretty interesting lab to work through. Even though I have used FTK in the past, I never did the full installation from scratch, so that definitely was the most difficult part. But after watching a youtube video and reading a couple of articles, I was able to get it working, and it was a piece of cake from there. Through this lab, I learned more how to recognize where to find certain artifacts, such as if I need to bash history, authentication logging, screenshots, scripts, etc. Going through this lab has added to my skillset of digital forensics, and a continuation of practicing writing reports explaining what I found, how I discovered it, and the resolution.